

Decomposing the variance of consumption-share growth: size versus level of renegotiations

1 Object and notation

Let s_{it}^g denote spouse g 's share of private consumption within couple i at age t (for the wife, $s^w = c^w / (c^w + c^m)$; for the husband, $s^m = 1 - s^w$), and let

$$y_{it}^g \equiv \Delta \log s_{it}^g = \log s_{i,t+1}^g - \log s_{it}^g$$

be its log growth over married cells. The object of interest is the (uncentered) volatility $\mathbb{E}[(y^g)^2]$, the “share component” of individual private-consumption volatility in the policy-experiment figures. In the model, s^g is a monotone function of the Pareto weight alone, so y^g is (up to numerical interpolation error) nonzero only in periods in which the weight is renegotiated. Let $\mathcal{R}$ denote the set of renegotiation cells (the bargaining weight differs from its lag) and $f \equiv \Pr(\mathcal{R})$ the renegotiation frequency.

2 Derivation

Write the log change as the product of the share change in *levels* and an effective inverse share:

$$y_{it}^g = \Delta S_{it}^g \cdot M_{it}^g, \quad \Delta S_{it}^g \equiv s_{i,t+1}^g - s_{it}^g, \quad M_{it}^g \equiv \frac{\Delta \log s_{it}^g}{\Delta S_{it}^g}.$$

This is an identity, not an approximation. By the mean value theorem applied to $\log(\cdot)$, $M_{it}^g = 1/\tilde{s}_{it}^g$ for some $\tilde{s}_{it}^g$ between s_{it}^g and $s_{i,t+1}^g$: M is the exact “1/share” *at which the change is evaluated*. Splitting $\mathbb{E}[(y^g)^2]$ into renegotiation and non-renegotiation cells and factoring the renegotiation part,

$$\mathbb{E}[(y^g)^2] = \underbrace{f \cdot \mathbb{E}[\Delta S^2 | \mathcal{R}] \cdot \mathbb{E}[(M^g)^2 | \mathcal{R}] \cdot R^g}_{\text{renegotiation part}} + \underbrace{(1-f) \mathbb{E}[(y^g)^2 | \neg \mathcal{R}]}_{\text{numerical noise (reported, small)}}, \quad (1)$$

where

$$R^g \equiv \frac{\mathbb{E}[\Delta S^2 (M^g)^2 | \mathcal{R}]}{\mathbb{E}[\Delta S^2 | \mathcal{R}] \mathbb{E}[(M^g)^2 | \mathcal{R}]}$$

is the correction that turns the product of averages into the average of the product. Equation (1) is exact; the four factors of the renegotiation part are log-additive, so changes across policy variants can be attributed factor by factor.

A structural implication built into the decomposition: since $\Delta S^m = -\Delta S^w$, the frequency f and the size factor $\mathbb{E}[\Delta S^2 | \mathcal{R}]$ are *identical for the two spouses*. Any difference between the wife's and the husband's share volatility—in level or in its response to policy—must run through the level factor $\mathbb{E}[(M^g)^2 | \mathcal{R}]$ and the interaction R^g .

3 Interpretation of the factors

f : **how often** the bargaining weight moves. A policy that tightens or relaxes participation constraints shows up here.

$\mathbb{E}[\Delta S^2 \mid \mathcal{R}]$: **how large** renegotiations are in *levels* of the share—bargaining risk with the proportional (log) scaling stripped out. Common to both spouses.

$\mathbb{E}[(M^g)^2 \mid \mathcal{R}]$: **where in the share distribution** the events land. A given level change is proportionally large for the spouse whose share is small: for a share near 0.3 this factor is roughly $(1/0.3)^2 \approx 11$, against $(1/0.7)^2 \approx 2$ for the other spouse. Because it is computed event by event, it captures composition: if a policy relocates renegotiations toward couples with more equal shares, this factor falls for the low-share spouse even if the mean share moves little.

R^g : the **pairing of size and location**. $R > 1$ means the large share changes happen precisely where the spouse’s share is small (damage compounds); $R < 1$ means they happen where the share is large (damage cushioned); $R = 1$ means size and location are unrelated and the size and level factors can be discussed independently. R is a bookkeeping term, not a separate economic channel: if it is close to one, it should not be interpreted.

The non-renegotiation term in (1) collects the microscopic share movements that occur off renegotiation cells (interpolation of the intra-period allocation over the weight/resources grids). It is reported as a share of $\mathbb{E}[(y^g)^2]$ and should be negligible; a non-negligible value would indicate that measured share volatility partly reflects numerical noise rather than bargaining.

4 Implementation

The decomposition is computed by `share_var_decomposition()` in `reg_cons_insurance.py` and printed at the end of each policy experiment (`alimony_experiment.py`, `WLP_experiment.py`, `tax_experiment.py`), one row per policy variant and spouse, on the same married cells as the corresponding volatility figures. Column mapping: $\mathbf{E[y2]x100} = \mathbb{E}[(y^g)^2] \times 100$; `freq` = f ; $\mathbf{E[dS2|r]x100} = \mathbb{E}[\Delta S^2 \mid \mathcal{R}] \times 100$; $\mathbf{E[M2|r]} = \mathbb{E}[(M^g)^2 \mid \mathcal{R}]$; $\mathbf{R} = R^g$; `nonren%` = the non-renegotiation share of $\mathbb{E}[(y^g)^2]$ in percent. The exactness of (1) is verified numerically at every call (an `identity gap` warning is printed if it fails).